Matched legacy/repaired `s4_expr` rerun: run report

Observation-only run record. EXP-SEMAEV-93d5cc and its successors EXP-SEMAEV-88c46c and EXP-SEMAEV-6e9dcf; hypothesis H-SEMAEV-94dab2; goal GOAL-ECDLP-001; approvals DEC-20261018-a7a0fd, DEC-20261019-5b9240, DEC-20261019-ebf1dc. This report states what ran and what it recorded. It gives no verdict, support or weakening; the evidence review (TASK-20261018-83a561, then a decision record) does that.

What ran

Every archived run whose code path calls `harness.semaev.s4_expr` was rerun twice, concurrently, on one host: arm **legacy** with `s4_expr` from the exact `semaev.py` bytes all those runs recorded (sha256 `53a23952…`, vendored), arm **repaired** with the current `harness/semaev.py` (sha256 `c417f194…`). Nothing else was substituted; programs, frozen instances, targets, seeds and caps are the archived ones. Host: this container, Linux x86_64, 4 CPUs, CPython 3.13, sympy 1.14.0, Singular 4.4.1. The archive ran on macOS arm64 (MTIC, FIB) and with Singular 4.3.2 (ICINV). Supervisor: `tools/experiment_execution.py`; checker `implementation/check.py` (arm fingerprints, member hashes, re-extraction). Commits are each run manifest's `code.commit`; every manifest records `dirty: false`.

Cell	Run	Archived run	Status	Wall s
icinv-m4	RUN-SEMAEV-9cab7b	RUN-ICINV-e0cd8f-m3class (m4)	validated	2098
fib-004	RUN-SEMAEV-3e69b6	RUN-FIB-004	validated	1816
fib-005	RUN-SEMAEV-4be606	RUN-FIB-005	validated	1808
fib-decision	RUN-SEMAEV-894e27	RUN-FIB-006C / 007 / 008	validated	9
mtic-007	RUN-SEMAEV-2a8498	RUN-MTIC-007	validated	1511
mtic-008	RUN-SEMAEV-fcd67c	RUN-MTIC-008	validated	1575
mtic-010b	RUN-SEMAEV-500a81	RUN-MTIC-010b	interrupted	—
mtic-010b	RUN-SEMAEV-980e8c (88c46c)	RUN-MTIC-010b	interrupted	—
mtic-010b	RUN-SEMAEV-0e5876 (6e9dcf)	RUN-MTIC-010b	validated	1516

The two interrupted attempts were killed by container restarts (00:37 and 01:45 UTC on 2026-10-19), 13 and 16 minutes in, before any aggregation; both are preserved as written, with no receipt, and are not results (amendments `AMD-EXP-SEMAEV-93d5cc-20261019-mtic010b-successor`, `AMD-EXP-SEMAEV-88c46c-20261019-successor`). Excluded by import analysis (never call `s4_expr`): RUN-MTIC-001..006, −009; RUN-FIB-001..003. RUN-MTIC-010 and RUN-FIB-006/006B are superseded in the archive by −010b and −006C.

Controls

CTRL-LEGACY-REPRO (legacy rerun against the archive, frozen rules in `compare.py`): passed in every completed cell. ICINV 552/552 m4 records equal; FIB per-target (capped, basis size, basis degree, trivial) equal on all 200 + 75 (C16) and 75 + 15 (C20) common targets, `n_measured` differences 0; MTIC per-target tuples equal on all 25, 13 and 12 attempted targets, attempted-count differences 0. mtic-010b (RUN-SEMAEV-0e5876): regime, K* and uncensored counts equal at 16, 20 and 24 bits, and K* and capped pattern equal for both ablation bases (B = 52: 13/13 capped, B = 115: 12/12). **CTRL-ARM-FINGERPRINT** and **CTRL-REEXTRACT** passed in every completed trial (check.py exit 0): legacy S4 has free symbols {x1, x2, x4}, total degree 16, 227 terms; repaired {x1, x2, x3, x4}, degree 12, 439 terms, residue 0 on the EXP-SEMAEV-a097b5 witness; every FIB spawn child logged its arm's module and source hash.

Recorded values (archived / legacy rerun / repaired rerun)

Cell	Quantity (measured)	Archived	Legacy	Repaired
icinv-m4	m4 `s4_support` values (records)	227 ×546, 225 ×4, 210 ×2	same	439 ×550, 391 ×2
icinv-m4	m4 singular locus dim/degree	2/60 ×552	same	2/39 ×552
icinv-m4	gauge m4 agreement	276/276	276/276	276/276
fib-004	C16-1 solved / capped of 200	200 / 0	200 / 0	200 / 0
fib-004	C16-1 median Groebner s	0.0279	0.0579	3.678
fib-004	C16-1 basis size (counts)	3: 16, 4: 184	same	14–19 (5 values)
fib-004	C16-2 measured (all capped), stop	75, curve cap	75, curve cap	31, run cap
fib-005	C20-1, C20-2 measured (all capped)	75, 15	75, 15	75, 15
fib-decision	C16-1 Spearman rho_s (p)	−0.024 (0.742)	−0.097 (0.171)	−0.366 (< 1e−4)
fib-decision	curves passing the gate; gate	none; fail	none; fail	none; fail
fib-decision	C16-1 predictor-cost ratio	8.6e−6	4.2e−6	6.5e−8
mtic-007	16-bit attempted / capped / cancelled	25/25/25	25/25/25	25/25/25

mtic-008	20-bit; 24-bit attempted / capped	13/13; 12/12	same	same
mtic-010b	regime, K* (all accountings)	(i), infinite	(i), infinite	(i), infinite

Derived values: medians are of the recorded per-target `cost_seconds`; the C16-1 median ratio repaired/legacy is 3.678 / 0.0579 ≈ 63.5 (legacy/archived ≈ 2.07, host speed). The p-values are the recorded 10,000-replicate permutation values (p reported as 0.0 for the repaired arm). The legacy-rerun versus archive spread of rho_s on identical code (0.073) is the recorded timing-noise A/A for that statistic; the repaired-versus-legacy difference is 0.269.

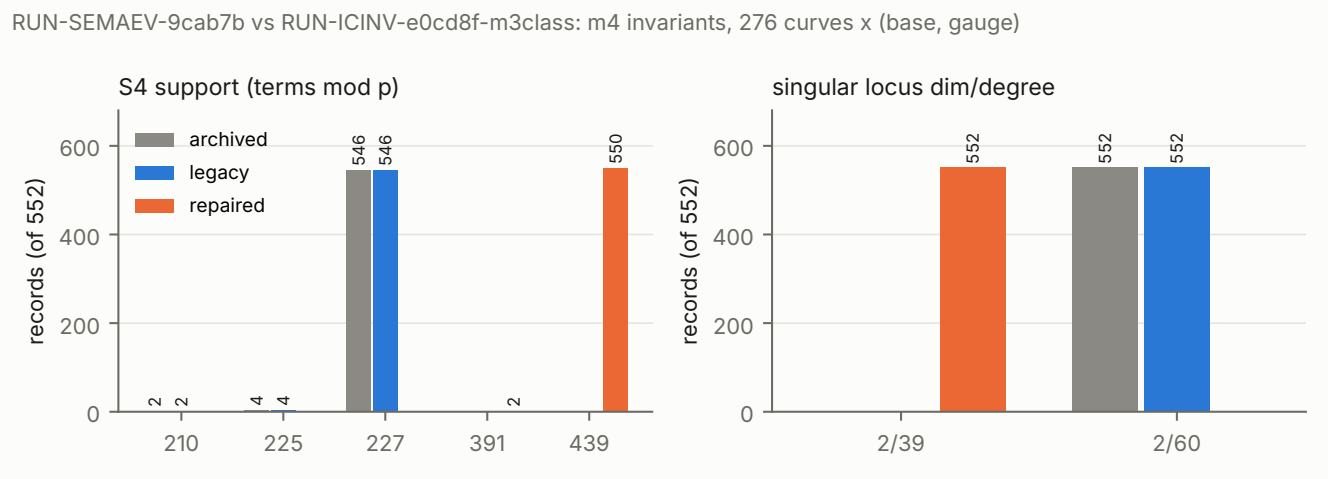


Figure 1: ICINV m4 invariants per record (552 = 276 curves × base and gauge twin) for RUN-ICINV-e0cd8f-m3class (archived) and RUN-SEMAEV-9cab7b (both arms). Counts are recorded values.

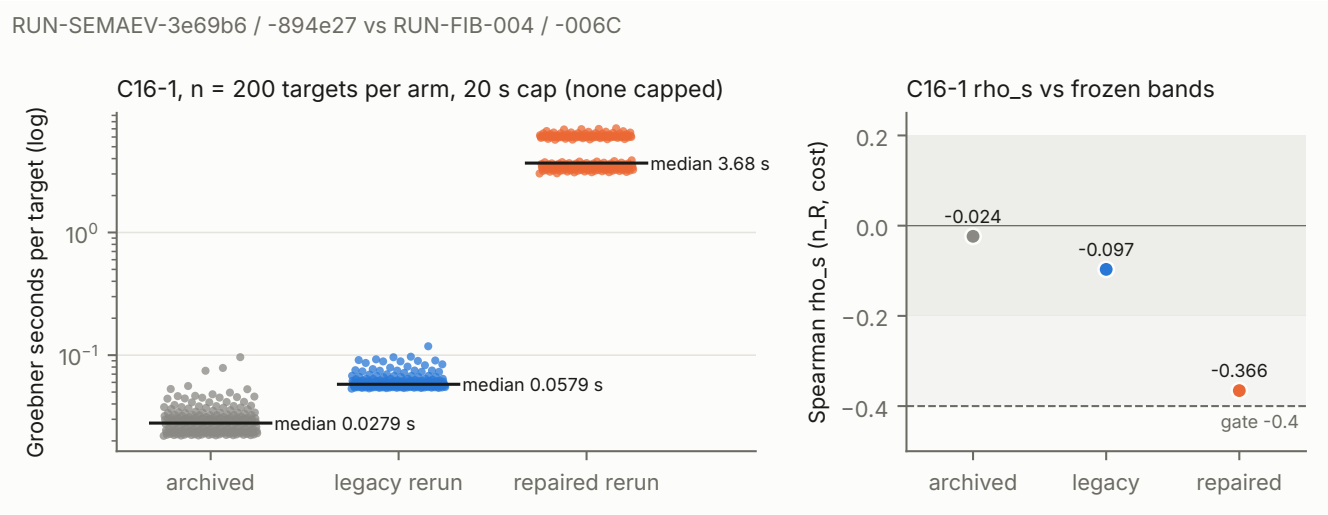


Figure 2: Left: recorded Groebner seconds per C16-1 target, n = 200 per arm (RUN-FIB-004; RUN-SEMAEV-3e69b6), log scale, bar = median. Right: recorded C16-1 Spearman rho_s (RUN-FIB-006C; RUN-SEMAEV-894e27) against the frozen EXP-FIB-001 bands: |rho| < 0.2 falsification band, 0.2–0.4 inconclusive, gate at −0.4.

Frozen conclusion fields (compare.conclusion_fields)

Repaired arm differs from the archive in: icinv-m4 `s4_support` value set and singular-locus value set (gauge agreement and the class-versus-control comparison unchanged). Equal to the archive in every other frozen field: FIB `resource_exhaustion` and all-capped flags for all four curves; correlation gate, curves passing, per-curve status, cost gate, gate and RUN-FIB-008 outcome; MTIC all-capped and median-at-cap flags at 16, 20 and 24 bits; MTIC-010b regime and K* per accounting at 16, 20 and 24 bits and K* for both ablation bases. Outside the frozen fields (recorded, not pre-registered as conclusion fields): the C16-1 rho_s moves from the EXP-FIB-001 falsification band (|rho| < 0.2) into its inconclusive band (0.2 ≤ |rho| < 0.4), and C16-2 measures 31 instead of 75 targets because the slower C16-1 solves consume the 1800 s run cap.

Limits

Toy tier only. Wall-clock caps make every capped count host-dependent; they reproduced exactly here, which is a property of this host. No cost, speedup or ECDLP claim; no hypothesis status changes. Sources: each run's `manifest.yaml`, `raw-result.json`, `arms.tar.gz` (member hashes in `arms-manifest.json`) and `execution-receipt.json`; `plot-data.json` and `chart.py` here reproduce both figures.